Sortify - Data Preprocessing Report

YOLOv11 Waste-Detection Dataset

1. Overview

Sortify needs a model that can recognise common waste items in images and connect each detection to one of five disposal groups: **Recyclable, General Waste, Organic, E-Waste,** or **Hazardous**. To prepare the dataset, five public datasets were combined, suitable object types were selected, and their labels were converted into one consistent format.

The original sources contained **13,108 images** and **56,162 labelled objects**. After filtering and cleaning, the final dataset contains **7,355 unique images** and **21,886 labelled objects** across **10 detector classes**.

2. Source Datasets

The sources cover different waste types and collection settings. Their labels and image formats were checked before combining them. The table below summarises the main classes in each original dataset; the full source class lists are longer, especially for MedBin and PackWISE.

Source dataset	Image type / setting	Raw images	Raw labelled objects	Major original classes	Source
Bangladesh Recyclable Waste	Outdoor and general waste	2,842	6,076	Paper, Plastic Bottle, Plastic bag, Plastic waste	Roboflow Universe
MedBin Dataset 2	Medical and clinical waste	1,796	6,410	Medical gloves, masks, N95 masks, syringes, needles, blades, scalpels, drug packaging, glass bottles, plastic medical bottles and bags, paper boxes, thermometers	Roboflow Universe · GitHub

Source dataset	Image type / setting	Raw images	Raw labelled objects	Major original classes	Source
Mendeley Trash Detection	Household waste and catalog-style images	5,049	34,082	Biodegradable, Cardboard, Glass, Metal, Paper, Plastic	Mendeley Data
PackWISE Packaging Waste	High-resolution industrial sorting belt	586	6,713	Plastic foil/bags, paper bags and other paper, metal cans and foil, plastic bottles, tissues, plastic cups/lids, blister packs, foam and other packaging	Fraunhofer Fordatis
RecyBat24	Batteries photographed from multiple views	2,835	2,881	Cylindrical, pouch, and prismatic batteries	Zenodo
Total raw pool	**Five source datasets**	**13,108**	**56,162**	**73 source classes in total**	—

Note: The Bangladesh source includes alternate partitions of the same underlying images. They were not counted as a separate dataset.

3. Preparing the Labels

Each image needs a label file that identifies its objects and gives the location of each object. The sources used different formats, so the labels were standardised for YOLOv11.

- **Converted shape outlines to boxes:** Some annotations traced the object's outline rather than drawing a rectangle. These were converted to rectangular bounding boxes.
- **Converted JSON annotations:** PackWISE and RecyBat24 stored their labels in JSON files. Their object locations and class IDs were converted into YOLO label files.
- **Resized large images:** PackWISE images were reduced from **4096 × 4096** to **1280 × 1280 pixels** to make them easier to load during training.
- **Removed duplicate or unmatched files:** Identified duplicate images and label files without matching images were removed.
- **Checked box coordinates:** Labels were checked for the required five values, valid class IDs, positive box sizes, and coordinates within the image.

4. Selecting and Combining Classes

The five sources originally contained **73 classes**. These were reduced to **10 classes** chosen for Sortify. The aim was to combine similar items without assigning objects to misleading labels.

- **Removed broad labels:** Labels such as `Plastic waste` and `PLASTIC` could include bottles, bags, cups, or wrappers. They were excluded because Sortify treats plastic bottles and plastic bags as separate classes.
- **Removed unclear or unsuitable labels:** Some labels described mixed materials, unclear fragments, specialised laboratory equipment, or examples unsuitable for the planned classes.
- **Removed affected images where needed:** If an image contained an object whose label was being excluded, the whole image was removed in the relevant cases. Otherwise, the model could see an object without being told what it was.
- **Filtered unusual examples:** Very large boxes covering nearly an entire image and dense produce collages were removed from selected Mendeley classes.
- **Limited repeated battery views:** RecyBat24 includes multiple photographs of the same physical battery. The number of views was limited, and all views of one battery were kept in the same data split.

These steps were intended to make the labels more consistent and reduce the chance of the model learning from contradictory examples.

5. Which Source Classes Were Used for Each Sortify Class?

This table shows which original dataset labels contributed to each final Sortify detector class. Some source classes were combined because they describe similar objects. Other source classes were left out because they were too broad, unclear, or did not fit the final classes.

Final Sortify class	Source dataset	Original class labels used
`paper`	Mendeley Trash Detection	`PAPER`
`paper`	PackWISE	`paper-bag`, `paper-other`, `paper-cup/pot`
`cardboard`	Mendeley Trash Detection	`CARDBOARD`
`cardboard`	MedBin Dataset 2	`paperbox`, `covid_buffer_box`

Final Sortify class	Source dataset	Original class labels used
`glass`	Mendeley Trash Detection	`GLASS`
`glass`	MedBin Dataset 2	`glass_bottle`
`metal`	Mendeley Trash Detection	`METAL`
`metal`	PackWISE	`metal-can`, `metal-foil`
`metal`	MedBin Dataset 2	`Oxygen_cylinder`
`plastic_bottle`	Bangladesh Recyclable Waste	`Plastic Bottle`
`plastic_bottle`	PackWISE	`plastic-bottle`
`plastic_bottle`	MedBin Dataset 2	`plastic_medical_bottle`
`plastic_bag`	Bangladesh Recyclable Waste	`Plastic bag`
`plastic_bag`	PackWISE	`plastic-foil/bag`
`plastic_bag`	MedBin Dataset 2	`plastic_medical_bag`
`sanitary_pe_waste`	PackWISE	`tissue`
`sanitary_pe_waste`	MedBin Dataset 2	`Mask`, `N95`, `medical_gloves`, `cotton_swab`, `iodine_swab`, `gauze`, `bandage`
`organic`	Mendeley Trash Detection	`BIODEGRADABLE` (after filtering dense produce-collage images)

Final Sortify class	Source dataset	Original class labels used
`battery_ewaste`	RecyBat24	`cylindric`, `pouch`, `prismatic`
`battery_ewaste`	MedBin Dataset 2	`Electronic_thermometer`
`hazardous_medical`	MedBin Dataset 2	`drug_packaging`, `syringe`, `Bloody_objects`, `unguent`, `mercury_thermometer`, `blade`, `scalpel`, `needle`
`hazardous_medical`	PackWISE	`blister`

Important exclusions

Some source classes were deliberately not used in the final dataset:

- **Bangladesh Recyclable Waste:** `Paper` and `Plastic waste` were excluded because they combine different object types that Sortify treats as separate classes.
- **Mendeley Trash Detection:** `PLASTIC` was excluded because it mixes different plastic items; some unusually large paper/cardboard boxes and dense organic collages were also filtered.
- **PackWISE:** unclear or mixed classes such as `undefined`, `aluminium-coated-foil`, `beverage-carton`, and several small or ambiguous packaging classes were excluded.
- **MedBin Dataset 2:** specialised laboratory equipment and selected small classes, including loose pills/capsules and the dataset's `Radioactive_objects` examples, were excluded.
- **RecyBat24:** the battery shape labels were combined into one `battery_ewaste` class. Repeated views were limited, and photos of the same physical battery were kept in one split.

The final instance counts in Section 6 are the counts after filtering, merging, and splitting. Source-class counts may not add up exactly to the final class total because images and annotations were removed during cleaning and some counts were measured at different stages.

6. Final Classes and Distribution

The model predicts one of the 10 detector classes below. Sortify's backend maps each class to a default disposal group. The correct disposal route can still depend on local rules and whether an item is contaminated.

ID	Detector class	Backend disposal group	Train boxes	Validation boxes	Test boxes	Total boxes	Images containing class
0	paper	Recyclable	1,967	272	257	**2,496**	879
1	cardboard	Recyclable	2,391	244	228	**2,863**	1,013
2	glass	Recyclable	1,653	232	318	**2,203**	860
3	metal	Recyclable	1,989	282	204	**2,475**	915
4	plastic_bottle	Recyclable	1,355	177	185	**1,717**	1,081
5	plastic_bag	General Waste	2,348	315	309	**2,972**	1,049
6	sanitary_pe_waste	General Waste	1,431	171	186	**1,788**	1,015
7	organic	Organic	1,990	276	180	**2,446**	618
8	battery_ewaste	E-Waste	1,184	139	159	**1,482**	1,446
9	hazardous_medical	Hazardous	1,166	136	142	**1,444**	983
—	**TOTAL**	**Five groups**	**17,474**	**2,244**	**2,168**	**21,886**	**7,355 unique images**

Note: "Images containing class" counts images that include that class. A single image can contain more than one class, so these per-class image counts should not be added together to calculate the number of unique images.

7. Training, Validation, and Test Sets

Split	Images	Labelled objects	Purpose
Training	5,887	17,474	Used to train the model to recognise the 10 classes.

Split	Images	Labelled objects	Purpose
Validation	734	2,244	Used during development to compare training runs and choose settings.
Test	734	2,168	Kept separate for the final evaluation after model choices are made.
Total	**7,355**	**21,886**	All unique images and labelled objects.

The split is approximately **80% training, 10% validation, and 10% testing**. Battery images were split by physical battery ID, so different photographs of the same battery do not appear in more than one split.

8. Verification Results

The final dataset passed the following structural checks:

- **Image and label matching:** Every final image has a matching label file; no unmatched images or labels were found.
- **Label format:** All checked object labels contain the required five values: class ID, horizontal centre, vertical centre, width, and height.
- **Bounding boxes:** No malformed label lines or boxes extending outside the image were found.
- **Class IDs:** Labels use the expected class IDs, from 0 to 9.
- **Duplicate image stems:** No duplicate Roboflow image stems were found across the audited splits.
- **Battery separation:** No physical battery ID is shared between the training, validation, and test sets.

These checks show that the files are consistently organised and that the listed split checks passed. They do **not** prove that every object is labelled correctly or that the trained model will perform well on real-world images. Model performance must be measured after training.

9. Summary

The final Sortify dataset combines five sources into one YOLOv11-compatible dataset with **7,355 images, 21,886 labelled objects, and 10 detector classes**. Labels were standardised, unsuitable examples were filtered, and the final files passed checks for matching labels, valid bounding boxes, and separation between the data splits.